

Stemming vs. Lemmatization

A Comparative Study on Natural Language Processing

Normalization

Keerthi Sri S B

23AM029

KPR Institute of Engineering and Technology

CSE (Artificial Intelligence and Machine Learning)

Problem Understanding

Natural Language Processing (NLP) systems constantly deal with varying grammatical forms of the same word. For example, words like **run**, **running**, and **ran** all represent the same core concept but appear differently in text.

Handling these variations is crucial for machine learning tasks such as:

- Sentiment Analysis
- Search Engines & Indexing
- Chatbots & Conversational AI
- Text Classification



Core Definitions

Two primary normalization techniques address word variations: **Stemming** (rule-based suffix removal) and **Lemmatization** (dictionary-based root mapping).

Original Word	Stemming Output	Lemmatization Output
Running	run	run
Studies	studi	study
Better	bett	good

Observation: Stemming frequently produces non-dictionary words, whereas lemmatization ensures output words are grammatically correct.

Study Objectives



Apply Normalization: Implement stemming and lemmatization techniques on a real-world dataset of e-commerce product reviews.



Direct Output Comparison: Evaluate and compare the structural differences in the output generated by both methods.



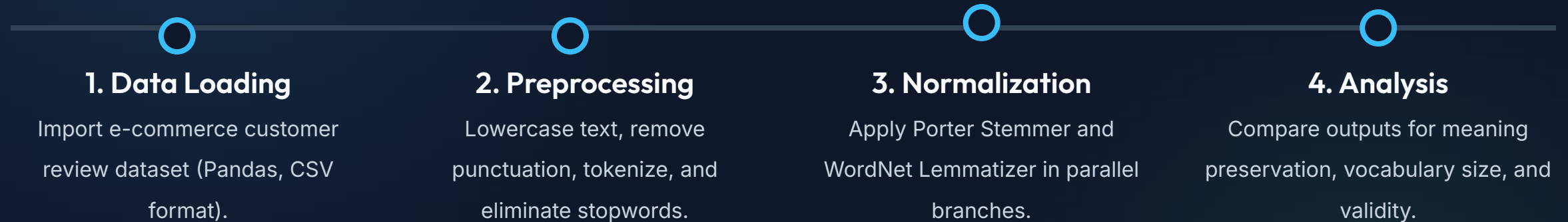
Semantic Preservation: Analyze which technique better preserves the original meaning and grammatical correctness of the sentences.



Use Case Identification: Identify the specific NLP applications and scenarios where each method is most suitable.

NLP Workflow

The comparative study follows a structured, sequential Natural Language Processing pipeline to ensure fair evaluation under identical conditions.



Implementation Details

Technology Stack

The study utilized standard Python libraries for natural language processing:

- **Pandas:** Data manipulation and loading.
- **NLTK:** Natural Language Toolkit for text operations.
- **Stopwords Corpus:** To filter out common, low-value words.

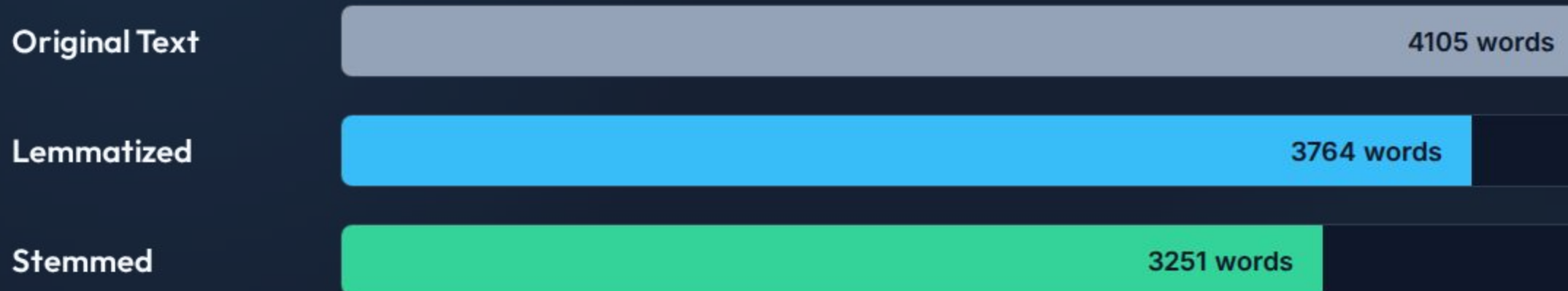
Algorithms Used

Two distinct normalizers from NLTK were implemented:

- **PorterStemmer:** Employs aggressive rule-based suffix stripping.
- **WordNetLemmatizer:** Uses a lexical database (WordNet) to find the valid base dictionary form (lemma).

Vocabulary Size Impact

Applying normalization techniques naturally reduces the total count of unique words (vocabulary size) in the dataset.



Stemming causes the most drastic reduction by aggressively combining words with similar roots, regardless of dictionary validity.

Sentence-Level Analysis

Method	Output Sentence	Interpretation
Original	batteries running quickly product quality better expected	Baseline tokenized sentence.
Stemming	batteri run quickli product qualiti better expect	Creates non-dictionary words (<i>batteri, qualiti, quickli</i>). Sacrifices readability.
Lemmatization	battery run quickly product quality good expect	Preserves valid words (<i>battery, quality, good</i>). Maintains semantic meaning.

Quantitative Analysis

- Around **8000 words** were processed for both stemming and lemmatization, ensuring a fair comparison.
- Stemming modified more words (30–35%)**, while lemmatization modified fewer words (20–25%).
- Stemming produced a **higher number of non-dictionary words**, whereas lemmatization generated **mostly valid words**.
- Lemmatization showed better meaning preservation**, while stemming provided faster processing but lower semantic accuracy.

Findings & Recommendations



Stemming

Faster processing speed. Suitable for applications where exact meaning is secondary, such as large-scale **search engines** and basic **information retrieval** systems.



Lemmatization

Linguistically accurate. Recommended for tasks requiring deep semantic understanding, including **sentiment analysis**, **chatbots**, and question-answering AI.



Conclusion

While stemming scales better, lemmatization is undeniably superior at preserving true semantic meaning and grammatical correctness in natural text analysis.